Numerical Methods: Full Derivation

Finite Difference Discretisation, Convection Parameterisation, and Bayesian Constraining

From Continuous Equations to Matrix Coefficients

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The Heat Equation in Conservative Form

Continuous form

The temperature field $T(z,t)$ in a radially stratified ice shell satisfies the energy equation with volumetric tidal heating:

$$\rho(T)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{1}{\mathcal{G}(z)}\, \frac{\partial}{\partial z}\left[\mathcal{G}(z)\, k(T)\, \frac{\partial T}{\partial z}\right] + q_{\text{tidal}}(T) \tag{1}$$

where z is depth from the surface, and the geometric factor accounts for coordinate system:

$$\mathcal{G}(z) = \begin{cases} 1 & \text{Cartesian (thin shell, } H < 30 \text{ km)} \\ r(z)^2 & \text{spherical (thick shell, } H \geq 30 \text{ km)} \end{cases} \tag{2}$$

with $r(z) = R_p - z$ the radial distance from Europa's centre.

The equation is in **flux-conservative form**: the divergence operator $\frac{1}{\mathcal{G}}\frac{\partial}{\partial z}(\mathcal{G}\, k\, \frac{\partial T}{\partial z})$ ensures that the discrete heat flux is conserved across cell interfaces, even when k jumps by orders of magnitude at the conductive–convective boundary.

Boundary conditions

- **Surface** ($z = 0$): fixed temperature $T(0,t) = T_{\text{surf}}$ (Dirichlet).
- **Base** ($z = H(t)$): pressure-dependent melting point $T(H,t) = T_m(H) = 273.0 - 7.4\times10^{-8}\, \rho\, g\, H$ (Dirichlet).

The shell thickness $H(t)$ evolves via the Stefan condition:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \frac{q_{\text{cond}}(H) - q_{\text{ocean}}}{\rho\, L} \tag{3}$$

Initial condition

The temperature is initialised as a linear profile:

$$T(z,0) = T_{\text{surf}} + \frac{T_m(H_0) - T_{\text{surf}}}{H_0}\, z \tag{4}$$

with initial shell thickness $H_0 = 10$ km.

Spatial Discretisation

Uniform grid

The domain $[0, H(t)]$ is divided into N equally spaced nodes $z_i = i\,\Delta z$ for $i = 0, 1, \ldots, N-1$, where:

$$\Delta z = \frac{H(t)}{N-1} \tag{5}$$

The grid uses normalised coordinates $\xi_i = i/(N-1) \in [0,1]$ so that physical positions $z_i = \xi_i \cdot H(t)$ automatically adjust when H changes. This avoids explicit remeshing.

Half-node conductivity: harmonic mean

At cell interfaces, the conductivity $k_{i+\frac{1}{2}}$ is computed as the **harmonic mean** of adjacent nodal values:

$$k_{i+\frac{1}{2}} = \frac{2\,k_i\,k_{i+1}}{k_i + k_{i+1}} \tag{6}$$

This choice is physically motivated: for heat flow through two resistances in series (as across a cell interface), the harmonic mean gives the correct effective conductivity. It is critical at the conductive–convective interface, where k may jump by a factor of Nu $\sim$ 10–20 when the Nusselt enhancement is applied. An arithmetic mean would overestimate the interface flux; the harmonic mean correctly limits it to the smaller of the two conductivities.

Geometric factors at half-nodes

In spherical geometry, the geometric factor at half-nodes is approximated by arithmetic averaging:

$$\mathcal{G}_{i+\frac{1}{2}} = \frac{\mathcal{G}_i + \mathcal{G}_{i+1}}{2} = \frac{r_i^2 + r_{i+1}^2}{2} \tag{7}$$

Discrete diffusion operator

Applying the conservative finite-difference stencil to the flux divergence term at interior node i:

$$\frac{1}{\mathcal{G}_i}\frac{\partial}{\partial z}\left[\mathcal{G}\,k\,\frac{\partial T}{\partial z}\right]_i \approx \frac{1}{\mathcal{G}_i\,\Delta z^2}\left[\mathcal{G}_{i+\frac{1}{2}}\,k_{i+\frac{1}{2}}\,(T_{i+1}-T_i) - \mathcal{G}_{i-\frac{1}{2}}\,k_{i-\frac{1}{2}}\,(T_i - T_{i-1})\right] \tag{8}$$

Define the shorthand coefficients:

$$F_i^+ = \frac{\mathcal{G}_{i+\frac{1}{2}}\,k_{i+\frac{1}{2}}}{\mathcal{G}_i\,\Delta z^2} \tag{9}$$

$$F_i^- = \frac{\mathcal{G}_{i-\frac{1}{2}}\,k_{i-\frac{1}{2}}}{\mathcal{G}_i\,\Delta z^2} \tag{10}$$

Then the discrete spatial operator is:

$$\mathcal{L}_i[T] = F_i^+\,(T_{i+1}-T_i) - F_i^-\,(T_i - T_{i-1}) = F_i^-\,T_{i-1} - (F_i^+ + F_i^-)\,T_i + F_i^+\,T_{i+1} \tag{11}$$

Time Integration: the θ-Method

General θ-method

The semi-discrete equation $\rho_i\,c_{p,i}\,\frac{\mathrm{d}T_i}{\mathrm{d}t} = \mathcal{L}_i[T] + q_i$ is integrated in time using the θ-weighted scheme:

$$\rho_i\,c_{p,i}\,\frac{T_i^{n+1} - T_i^n}{\Delta t_{\text{eff}}} = \theta\,\mathcal{L}_i[T^{n+1}] + (1-\theta)\,\mathcal{L}_i[T^n] + q_{\text{tidal},i} \tag{12}$$

- $\theta = 1$: backward Euler (first-order, L-stable, strongly dissipative).
- $\theta = \frac{1}{2}$: Crank–Nicolson (second-order, A-stable, minimally dissipative).

Define the local CFL-like ratio:

$$\alpha_i = \frac{\Delta t_{\text{eff}}}{\rho_i\,c_{p,i}} \tag{13}$$

Rearrangement into tridiagonal form

Substituting Eq. 11 into Eq. 12 and collecting terms on T^{n+1} (left-hand side) and T^n (right-hand side):

Left-hand side (implicit):

$$\underbrace{-\theta\,\alpha_i\,F_i^-}_{a_i}\;T_{i-1}^{n+1} + \underbrace{\left[1+\theta\,\alpha_i\,(F_i^+ + F_i^-)\right]}_{b_i}\;T_i^{n+1} + \underbrace{-\theta\,\alpha_i\,F_i^+}_{c_i}\;T_{i+1}^{n+1} \tag{14}$$

The tridiagonal coefficients are therefore:

$$\boxed{a_i = -\theta\,\alpha_i\,F_i^-} \quad \text{(sub-diagonal)} \tag{15}$$

$$\boxed{b_i = 1+\theta\,\alpha_i\,(F_i^+ + F_i^-)} \quad \text{(main diagonal)} \tag{16}$$

$$\boxed{c_i = -\theta\,\alpha_i\,F_i^+} \quad \text{(super-diagonal)} \tag{17}$$

Right-hand side (explicit + source):

$$d_i = T_i^n + (1-\theta)\,\alpha_i\left[F_i^{+,n}(T_{i+1}^n - T_i^n) - F_i^{-,n}(T_i^n - T_{i-1}^n)\right] + \alpha_i\,q_{\text{tidal},i} \tag{18}$$

where the superscript n on $F^{\pm,n}$ indicates that the explicit flux coefficients are evaluated at the old-time material properties. The tidal heating source term enters the right-hand side directly as:

$$S_i = \frac{\Delta t_{\text{eff}}\cdot q_{\text{tidal}}(T_i)}{\rho_i\,c_{p,i}} \tag{19}$$

Special case: backward Euler ($\theta = 1$)

When $\theta = 1$, the explicit flux term $(1-\theta)(\ldots)$ vanishes, and the right-hand side simplifies to:

$$d_i = T_i^n + S_i \qquad (\theta = 1) \tag{20}$$

This is computationally cheaper (no explicit flux evaluation) and is used during Rannacher startup.

Rannacher startup procedure

The Crank–Nicolson scheme ($\theta = \frac{1}{2}$) is second-order accurate but can produce spurious oscillations when applied to stiff initial conditions (e.g., a sharp temperature gradient that develops in early timesteps). The Rannacher startup (Rannacher 1984) stabilises the scheme by using backward Euler for the first few steps:

1. For the first $N_R = 4$ logical timesteps:
   (a) Set $\theta = 1$ and $\Delta t_{\text{eff}} = \Delta t/2$.
   (b) Assemble the tridiagonal system and solve for T^* (first half-step).
   (c) Using T^* as the new initial condition, assemble and solve again for T^{n+1} (second half-step).

   Each logical step therefore comprises two backward-Euler half-steps, advancing by a total of Δt.

2. For all subsequent steps ($n \geq N_R$):
   - Set $\theta = \frac{1}{2}$ and $\Delta t_{\text{eff}} = \Delta t$.
   - Assemble and solve the full Crank–Nicolson system.

The backward-Euler half-steps are unconditionally stable and strongly dissipate high-frequency oscillations, ensuring a smooth handoff to the Crank–Nicolson scheme.

Nonlinear Iteration: Picard Method

The material properties $k(T)$, $\rho(T)$, $c_p(T)$, and $q_{\text{tidal}}(T)$ all depend on temperature, making the system nonlinear. At each timestep, the following Picard (fixed-point) iteration is applied:

1. Set the initial guess $T^{(0)} = T^n$ (previous timestep solution).
2. For $m = 0, 1, 2$ (maximum 3 iterations):
   (a) Evaluate material properties at $T^{(m)}$: $k_i^{(m)} = k(T_i^{(m)})$, $\rho_i^{(m)} = \rho(T_i^{(m)})$, $c_{p,i}^{(m)} = c_p(T_i^{(m)})$, $q_i^{(m)} = q_{\text{tidal}}(T_i^{(m)})$.
   (b) Compute half-node conductivities $k_{i+\frac{1}{2}}^{(m)}$ via the harmonic mean (Eq. 6).
   (c) Assemble the tridiagonal system using coefficients Eqs. 15–17 and right-hand side Eq. 18, with implicit coefficients from $T^{(m)}$ and explicit fluxes from T^n.
   (d) Solve the tridiagonal system via the Thomas algorithm to obtain $T^{(m+1)}$.
   (e) Check convergence: if $\max_i |T_i^{(m+1)} - T_i^{(m)}| < 0.01\,\text{K}$, accept $T^{n+1} = T^{(m+1)}$ and exit.
3. If convergence is not achieved after 3 iterations, accept the last iterate as T^{n+1}.

In practice, convergence is typically achieved in 1–2 iterations because the temperature change per timestep is small relative to the absolute temperature.

Tridiagonal Solution: Thomas Algorithm

The assembled system $\mathbf{A}\,\mathbf{T}^{n+1} = \mathbf{d}$ is a tridiagonal matrix equation of dimension N:

$$\begin{pmatrix} b_0 & c_0 & & & & \\ a_1 & b_1 & c_1 & & & \\ & a_2 & b_2 & c_2 & & \\ & & \ddots & \ddots & \ddots & \\ & & & a_{N-2} & b_{N-2} & c_{N-2} \\ & & & & a_{N-1} & b_{N-1} \end{pmatrix} \begin{pmatrix} T_0 \\ T_1 \\ T_2 \\ \vdots \\ T_{N-2} \\ T_{N-1} \end{pmatrix} = \begin{pmatrix} d_0 \\ d_1 \\ d_2 \\ \vdots \\ d_{N-2} \\ d_{N-1} \end{pmatrix} \tag{21}$$

This is solved via `scipy.linalg.solve_banded` (the Thomas algorithm), which performs LU decomposition in $O(N)$ operations:

Forward sweep (eliminate sub-diagonal):

$$w_i = \frac{a_i}{b'_{i-1}} \tag{22}$$

$$b'_i = b_i - w_i\, c_{i-1} \tag{23}$$

$$d'_i = d_i - w_i\, d'_{i-1} \tag{24}$$

Back substitution:

$$T_{N-1} = \frac{d'_{N-1}}{b'_{N-1}} \tag{25}$$

$$T_i = \frac{d'_i - c_i\, T_{i+1}}{b'_i} \qquad i = N-2, \ldots, 0 \tag{26}$$

Boundary Condition Implementation

Surface node ($i = 0$): Dirichlet

For a fixed surface temperature, the first row of the tridiagonal system is:

$$\underbrace{1}_{b_0} T_0^{n+1} + \underbrace{0}_{c_0} T_1^{n+1} = \underbrace{T_{\text{surf}}}_{d_0} \tag{27}$$

Surface node ($i = 0$): Stefan–Boltzmann radiative

For the radiative boundary condition $\varepsilon\sigma T_s^4 = (1-a)Q + k\,\frac{T_1 - T_s}{\Delta z}$, the T^4 nonlinearity is linearised around the current temperature estimate $T_s^{(m)}$:

$$T^4 \approx 4\,(T_s^{(m)})^3\,T_s^{n+1} - 3\,(T_s^{(m)})^4 \tag{28}$$

Substituting and rearranging into the matrix row $b_0\,T_0^{n+1} + c_0\,T_1^{n+1} = d_0$:

$$b_0 = 4\,\varepsilon\sigma\,(T_s^{(m)})^3 + \frac{k_{\frac{1}{2}}}{\Delta z} \tag{29}$$

$$c_0 = -\frac{k_{\frac{1}{2}}}{\Delta z} \tag{30}$$

$$d_0 = (1-a)\,Q_\odot + 3\,\varepsilon\sigma\,(T_s^{(m)})^4 \tag{31}$$

where $k_{\frac{1}{2}}$ is the half-node conductivity between nodes 0 and 1.

Basal node ($i = N-1$): melting-point Dirichlet

The last row enforces the pressure-dependent melting temperature:

$$\underbrace{1}_{b_{N-1}} T_{N-1}^{n+1} = T_m(H) = 273.0 - 7.4 \times 10^{-8}\,\rho\,g\,H \tag{32}$$

Stefan condition: shell thickness evolution

After solving for T^{n+1}, the basal heat flux is computed via a second-order one-sided difference:

$$q_{\text{cond}} = k(T_{N-1})\,\frac{3\,T_{N-1} - 4\,T_{N-2} + T_{N-3}}{2\,\Delta z} \tag{33}$$

The freezing-front velocity is:

$$v = \frac{q_{\text{cond}} - q_{\text{ocean}}}{\rho\,L} \tag{34}$$

The shell thickness is updated explicitly:

$$H^{n+1} = \max(H^n + v\,\Delta t_{\text{eff}},\ 500\ \text{m}) \tag{35}$$

The 500 m floor prevents the shell from vanishing entirely, which would produce a singular grid. Equilibrium is declared when $|v| < 10^{-15}$ m/s.

Because the grid is defined on normalised coordinates $\xi_i \in [0, 1]$, the physical node positions $z_i = \xi_i\,H$ and spacing $\Delta z = H/(N-1)$ automatically adjust when H changes. No explicit remeshing or interpolation is required.

Convection Parameterisation: Full Algorithm

Physical basis

Europa's ice shell is expected to convect in the stagnant-lid regime: a cold, rigid conductive lid overlies a slowly overturning convective interior. The model does not resolve convective flow explicitly. Instead, it follows the parameterised approach of Green et al. (2021) and Deschamps & Vilella (2021), which replaces the convective heat transport with an enhanced effective conductivity.

The algorithm has four stages:

1. Compute the rheological transition temperature T_c.
2. Scan the temperature profile to find the interface depth.
3. Compute Ra and Nu for the convective sublayer.
4. Modify the conductivity profile.

Stage 1: Transition temperature T_c

The transition temperature marks where viscosity becomes low enough for convective participation.

Interior temperature (Deschamps & Vilella 2021, Eq. 18)

The mean temperature of the convective interior is:

$$T_i = B\left[\sqrt{1 + \frac{2}{B}\left(T_m - c_2\,\Delta T_s\right)} - 1\right] \tag{36}$$

where:

$$B = \frac{Q_v}{2\,R\,c_1}, \qquad \Delta T_s = T_m - T_{\text{surf}} \tag{37}$$

with $c_1 = 1.43$ and $c_2 = -0.03$ (Deschamps & Vilella 2021), $Q_v = 59{,}400$ J/mol, and $R = 8.314$ J/(mol·K).

Viscous temperature scale

The Frank–Kamenetskii parameter characterises the exponential viscosity contrast:

$$A = \frac{Q_v}{N\,R\,T_m} \tag{38}$$

where $N = 1$ for Newtonian (diffusion) creep. The viscosity at T_i and its derivative are:

$$\eta(T_i) = \eta_{\text{ref}}\,\exp\left[A\left(\frac{T_m}{T_i} - 1\right)\right] \tag{39}$$

$$\left.\frac{\mathrm{d}\eta}{\mathrm{d}T}\right|_{T_i} = -\eta(T_i)\,\frac{A\,T_m}{T_i^2} \tag{40}$$

The viscous temperature scale is:

$$\Delta T_v = -\frac{\eta(T_i)}{\left.\mathrm{d}\eta/\mathrm{d}T\right|_{T_i}} = \frac{T_i^2}{A\,T_m} = \frac{N\,R\,T_i^2}{Q_v} \tag{41}$$

Lid-base temperature

The conductive lid extends from the surface down to the depth where the temperature contrast from the convective interior equals θ_{lid} viscous temperature scales:

$$\boxed{T_c = T_i - \theta_{\text{lid}}\,\Delta T_v} \tag{42}$$

with $\theta_{\text{lid}} = 2.24$ (Green et al. 2021).

Alternative: Howell (2021) method

A simpler analytical estimate sets:

$$T_c^{\text{Howell}} = \frac{\sqrt{4\,T_m\,(R/Q_v) + 1} - 1}{2\,(R/Q_v)} \tag{43}$$

and the conductive base temperature as $T_{\text{cond,base}} = 2\,T_c - T_m$. This is faster but less accurate than the Green method for strongly internally heated shells.

Stage 2: Interface depth from temperature profile

Given T_c, the algorithm scans the numerical temperature profile $T_0, T_1, \ldots, T_{N-1}$ from the surface:

1. Find the first index j such that $T_j \geq T_c$.
2. If $0 < j < N-1$, linearly interpolate to find the exact interface depth:

$$z_c = z_{j-1} + \frac{T_c - T_{j-1}}{T_j - T_{j-1}}\,(z_j - z_{j-1}) \tag{44}$$

3. Set $D_{\text{cond}} = z_c$ and $D_{\text{conv}} = H - z_c$.

   If no node satisfies $T_j \geq T_c$, the entire shell is conductive ($D_{\text{conv}} = 0$, $\text{Nu} = 1$).

Stage 3: Rayleigh and Nusselt numbers

Rayleigh number

The Rayleigh number of the convective sublayer is:

$$\text{Ra} = \frac{\rho\,g\,\alpha_{th}\,\Delta T_{\text{conv}}\,D_{\text{conv}}^3}{\kappa\,\bar{\eta}} \tag{45}$$

where:

$$\Delta T_{\text{conv}} = T_m - T_c \tag{46}$$
$$\bar{T} = \tfrac{1}{2}(T_m + T_c) \tag{47}$$
$$\rho = \rho(\bar{T}) = 917\,[1 + 1.6 \times 10^{-4}\,(273 - \bar{T})] \tag{48}$$
$$k = k(\bar{T}) = 567/\bar{T} \tag{49}$$
$$c_p = c_p(\bar{T}) = 7.49\,\bar{T} + 90 \tag{50}$$
$$\kappa = k/(\rho\,c_p) \quad \text{(thermal diffusivity)} \tag{51}$$
$$\bar{\eta} = \eta(\bar{T}, d_{\text{grain}}) \quad \text{(composite viscosity, Eq. 67)} \tag{52}$$

Convection is active if and only if $\text{Ra} \geq \text{Ra}_{\text{crit}} = 1000$.

Nusselt number (Green et al. 2021)

For $\mathrm{Ra} \geq \mathrm{Ra}_{\mathrm{crit}}$:

$$\boxed{\mathrm{Nu} = C\,\mathrm{Ra}^{\xi}\left(\frac{T_i - T_c}{\Delta T_{\mathrm{conv}}}\right)^{\zeta}} \tag{53}$$

with $C = 0.3446$, $\xi = 1/3$ (Solomatov & Moresi 2000), and $\zeta = 4/3$ (internal heating correction; Green et al. 2021).

The ratio $(T_i - T_c)/\Delta T_{\mathrm{conv}}$ accounts for the fact that internal tidal heating raises the interior temperature T_i above what would be expected from purely basal heating. When $T_i > T_c$ (always true by construction), this ratio is > 0 and amplifies the Nusselt number. It is clipped to a minimum of 0.01 for numerical safety.

For $\mathrm{Ra} < \mathrm{Ra}_{\mathrm{crit}}$: $\mathrm{Nu} = 1$ (pure conduction).

Stage 4: Conductivity profile modification

The conductivity profile used in the tridiagonal assembly is modified as follows:

1. Start with the base conductivity at each node:

$$k_i^{\mathrm{base}} = \frac{567}{T_i} \tag{54}$$

2. Apply porosity reduction in the cold upper shell:

$$k_i = \begin{cases} k_i^{\mathrm{base}}\,(1 - f_{\mathrm{porosity}}) & T_i < T_\phi \\ k_i^{\mathrm{base}} & T_i \geq T_\phi \end{cases} \tag{55}$$

3. Apply the Nusselt enhancement below the conductive–convective interface. For all nodes at or below the interface index j_c:

$$k_i \leftarrow \mathrm{Nu}_{\mathrm{eff}} \cdot k_i \qquad \text{for } i \geq j_c \tag{56}$$

where:

$$\mathrm{Nu}_{\mathrm{eff}} = 1 + \alpha_{\mathrm{ramp}}\,(\mathrm{Nu} - 1) \tag{57}$$

4. Compute half-node conductivities via harmonic mean (Eq. 6) for the tridiagonal assembly.

Convection ramp for cold columns

For very cold surface conditions ($T_{\mathrm{surf}} \lesssim 60$ K, relevant for polar latitudes), the transition to convection may be numerically abrupt. A C^1-continuous smoothstep function ramps the Nusselt enhancement:

$$\alpha_{\mathrm{ramp}}(T_{\mathrm{surf}}) = \begin{cases} 0 & T_{\mathrm{surf}} \leq 60\ \mathrm{K} \\ 3t^2 - 2t^3 & 60 < T_{\mathrm{surf}} < 80\ \mathrm{K} \\ 1 & T_{\mathrm{surf}} \geq 80\ \mathrm{K} \end{cases} \tag{58}$$

where $t = (T_{\mathrm{surf}} - 60)/20$. This Hermite basis function has $\alpha_{\mathrm{ramp}}(60) = 0$, $\alpha_{\mathrm{ramp}}(80) = 1$, and zero derivative at both endpoints, ensuring smooth behaviour at latitude column boundaries.

Complete Timestep Algorithm

Each timestep proceeds as follows:

1. **Determine time integration parameters.** If $n < N_R = 4$: set $\theta = 1$, $\Delta t_{\mathrm{eff}} = \Delta t/2$. Otherwise: set $\theta = \frac{1}{2}$, $\Delta t_{\mathrm{eff}} = \Delta t$.

2. **Begin Picard loop** ($m = 0, 1, 2$):
   (a) Evaluate $k_i^{(m)}$, $\rho_i^{(m)}$, $c_{p,i}^{(m)}$, $q_{\text{tidal},i}^{(m)}$ at $T^{(m)}$.
   (b) **Build conductivity profile:**
   - Compute T_c via Green/Deschamps (Eqs. 36–42).
   - Scan profile for interface depth z_c (Eq. 44).
   - Compute Ra (Eq. 45) and Nu (Eq. 53).
   - Apply porosity and Nusselt modifications to k profile.
   - Compute harmonic-mean half-node conductivities.

   (c) **Assemble tridiagonal system:** compute a_i, b_i, c_i (Eqs. 15–17) and d_i (Eq. 18).
   (d) **Apply boundary conditions:** set row 0 (surface) and row $N-1$ (base).
   (e) **Solve** via Thomas algorithm $\rightarrow T^{(m+1)}$.
   (f) **Check convergence:** $\max_i |T_i^{(m+1)} - T_i^{(m)}| < 0.01$ K.
3. **If Rannacher step** ($n < N_R$): repeat step 2 for the second half-step using $T^{(m+1)}$ as the new initial condition.
4. **Update shell thickness** via Stefan condition (Eq. 3).
5. **Check equilibrium:** if $|\mathrm{d}H/\mathrm{d}t| < 10^{-15}$ m/s, halt.
6. Increment n and return to step 1.

Bayesian Importance Reweighting

Motivation

The forward model maps sampled parameters $\boldsymbol{\theta}_i$ to an equilibrium conductive lid thickness $D_{\text{cond},i}$. After running N Monte Carlo realisations, we wish to update the ensemble to reflect the Juno MWR observation $D_{\text{cond}}^{\text{obs}} = 29 \pm 10$ km without re-running the expensive forward model.

Likelihood

The likelihood of realisation i given the observation is:

$$\mathcal{L}_i = P(D^{\text{obs}} \mid D_{\text{cond},i}) = \frac{1}{\sqrt{2\pi}\,\sigma_{\text{tot}}} \exp\left[-\frac{1}{2}\left(\frac{D_{\text{cond},i} - D^{\text{obs}}}{\sigma_{\text{tot}}}\right)^2\right] \tag{59}$$

where:

$$\sigma_{\text{tot}} = \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{model}}^2} = \sqrt{10^2 + 3^2} \approx 10.4 \text{ km} \tag{60}$$

The model discrepancy $\sigma_{\text{model}} = 3$ km accounts for numerical resolution and parameterisation uncertainty.

Importance weights

Each realisation receives a normalised importance weight:

$$w_i = \frac{\mathcal{L}_i}{\sum_{j=1}^{N} \mathcal{L}_j} \tag{61}$$

In practice, to avoid numerical underflow, the log-weights are computed first:

$$\ln w_i = -\frac{1}{2}\left(\frac{D_{\text{cond},i} - D^{\text{obs}}}{\sigma_{\text{tot}}}\right)^2 \tag{62}$$

then stabilised by subtracting the maximum before exponentiating:

$$w_i = \frac{\exp(\ln w_i - \max_j \ln w_j)}{\sum_k \exp(\ln w_k - \max_j \ln w_j)} \tag{63}$$

Effective sample size

The quality of the reweighting is monitored via the effective sample size:

$$\mathrm{ESS} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2} = \frac{1}{\sum_i \tilde{w}_i^2} \tag{64}$$

where $\tilde{w}_i = w_i / \sum_j w_j$ are the normalised weights.

An ESS fraction $\mathrm{ESS}/N > 10\%$ indicates acceptable prior–likelihood overlap. If $\mathrm{ESS}/N < 10\%$, the prior is too diffuse relative to the likelihood, and the posterior is dominated by a few high-weight samples.

Weighted posterior statistics

Posterior percentiles are computed from the weighted empirical CDF. Sort the $D_{\mathrm{cond},i}$ values and their weights, compute the cumulative weight, and interpolate:

$$\hat{q}_p : \quad \sum_{j: D_j \le \hat{q}_p} w_j = p \tag{65}$$

The posterior median ($p = 0.5$), 68% interval ($p = 0.16, 0.84$), and 90% interval ($p = 0.05, 0.95$) are reported.

Iterative prior tightening

When the initial prior is broad, the posterior may be poorly resolved. The model implements an iterative scheme:

1. Run Monte Carlo with unconstrained (audited) priors.
2. Compute importance weights and posterior statistics.
3. Derive tightened priors from the posterior:
   - q_{basal}: new uniform bounds from posterior [5th, 95th] percentiles.
   - d_{grain}: new lognormal centre from posterior median, new σ from posterior 68% log-width.
4. Re-run Monte Carlo with tightened priors.
5. Repeat until the posterior median falls within the target range and $\mathrm{ESS}/N > 10\%$.

Bayes factors for model comparison

To compare ocean transport scenarios $\mathcal{M}_j$ and $\mathcal{M}_k$, the Bayes factor is:

$$\mathrm{BF}_{jk} = \frac{P(D^{\mathrm{obs}} \mid \mathcal{M}_j)}{P(D^{\mathrm{obs}} \mid \mathcal{M}_k)} = \frac{\frac{1}{N_j} \sum_i \mathcal{L}_i^{(j)}}{\frac{1}{N_k} \sum_i \mathcal{L}_i^{(k)}} \tag{66}$$

The marginal likelihood for each scenario is the mean likelihood across its ensemble, estimated by the arithmetic mean of the (unnormalised) weights.

Composite Viscosity Reference

For completeness, the composite diffusion-creep viscosity used in Ra computation and tidal heating:

$$\eta(T, d) = \frac{1}{2}\left[\frac{42\,\Omega}{R\,T\,d^2}\left(D_{0v}\,e^{-Q_v/(RT)} + \frac{\pi\,\delta}{d}\,D_{0b}\,e^{-Q_b/(RT)}\right)\right]^{-1} \tag{67}$$

clipped to $[10^{12},\ 10^{25}]$ Pa·s.